Topic 3: Production and Growth

Introduction to Economics II

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Based on Mankiw & Taylor, Chapters 22, 32 & 35

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Why Study Long-Run Growth?

The biggest question in economics

In Topic 1 we learned to **measure** economic activity. Now we ask **why**:

1. What determines the **quantity of goods and services** an economy can produce?
2. How is **national income distributed** between workers and capital owners?
3. Why are some countries **rich** and others **poor**?

Our tools: We will bring to the macroeconomic level two concepts from Intro I:

- The **production function** (how inputs become output)
- The **demand for factors** (how firms decide how much labor and capital to use)

Topics 2-3 focus on the **long run**: what determines production capacity over decades.

GDP Measures Prosperity

Real GDP per capita is the best available indicator of a country's material standard of living. Its growth rate tells us how fast prosperity is improving.

Five key facts about economic growth:

- **Fact 1:** Enormous variation in per capita income across countries. The poorest countries have per capita incomes less than 5% of those in the richest countries.
- **Fact 2:** Growth rates vary substantially across countries.
- **Fact 3:** Growth rates are not constant over time. For the world as a whole, growth rates were close to zero over most of history. For individual countries, growth rates also change over time.
- **Fact 4:** Relative position in the world distribution is not immutable. Countries can move from being poor to being rich (or vice versa).
- **Fact 5:** Per capita income is strongly correlated with welfare outcomes.

Why do these differences exist? That's what this topic is about.

Income Across the World

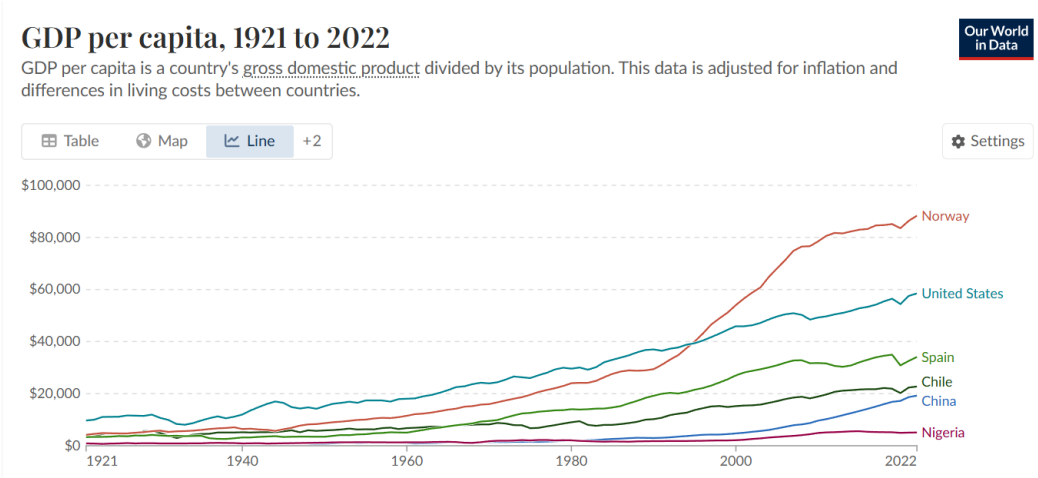


Figure: **Fact 1:** Enormous cross-country income differences

Growth Paths Vary Substantially

Figure: **Fact 2:** Different countries, different growth rates

The Historical Perspective

Key observation: For most of human history, living standards barely changed. Then, something remarkable happened:

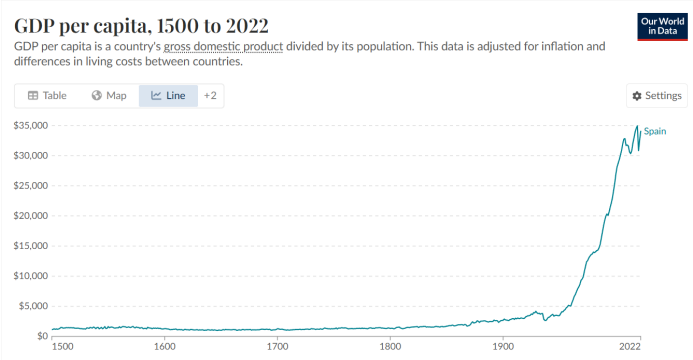


Figure: **Fact 3:** Growth is a recent phenomenon. *Source:* Our World in Data, Maddison Project Database

Fact 4: Relative Positions Are Not Immutable

Countries can move up or down in the world income distribution

Example 1: Argentina's decline

- Late 19th/early 20th century: Among the world's richest countries
- GDP per capita comparable to France and Germany
- Today: Middle-income country, far behind Western Europe

Example 2: China's rise

- 1960s: One of the world's poorest countries
- Since 1978 reforms: Sustained $\sim$9% average growth for 40 years
- Today: one of the largest-economies, approaching high-income status

GDP and Living Standards

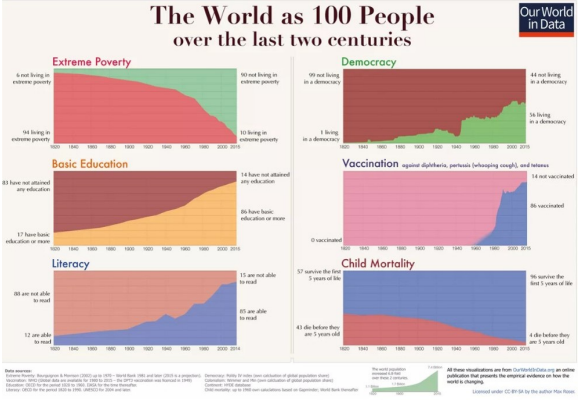


Figure: **Fact 5:** Higher GDP correlates with better welfare outcomes

Recall: Long Run vs Short Run (from Topic 1)

Long run

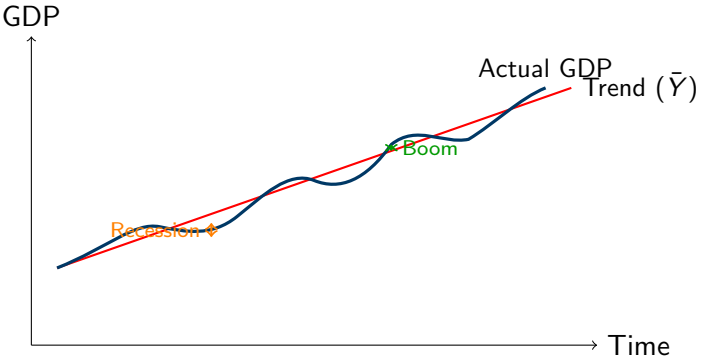
- All prices are flexible
- All markets clear
- Economy at full capacity
- Output determined by **supply side**

Short run

- Some prices are sticky
- Markets may not clear
- Economy above or below capacity
- Output also affected by **demand**

	This topic	**Topics 7-8**
Time horizon	Long run	Short run
Prices	Flexible	Sticky
Output set by	Supply	Supply + Demand

The Trend and the Cycle



From the data, we distinguish two phenomena:

The trend = Economic growth

Steady increase over decades.

Driven by technology, capital, labor.

The fluctuations = Business cycles

Output above/below trend.

Driven by specific shocks.

Potential GDP and the Output Gap

Two key concepts:

Potential GDP ($\bar{Y}$)

- What we **could** produce at full capacity
- Determined by A, K, L, H, N
- Smooth trend over time
- Also called: natural rate of output

Actual GDP (Y)

- What we **actually** produce
- Can be above or below $\bar{Y}$
- Fluctuates with business cycles
- Determined by AD in the short run

Output gap: $Y - \bar{Y}$

- $Y > \bar{Y}$: Economy overheating (inflationary pressure)
- $Y < \bar{Y}$: Recession (unused capacity, high unemployment)
- $Y = \bar{Y}$: Full employment (long-run equilibrium)

In the long run, $Y \to \bar{Y}$. This topic explains what determines $\bar{Y}$.

Why Is the Long-Run Supply Curve Vertical?

The key difference between micro and macro supply

In Microeconomics:

- If the price of bicycles rises (other prices constant)
- Firms earn higher profits → produce more bicycles
- Supply curve is **upward-sloping**

In Macroeconomics (long run):

- If the *general* price level rises, ALL prices rise
- Revenue rises, but so do costs (wages, materials, rent...)
- No incentive to change production!

The crucial distinction:

- **Relative price change** → Changes incentives → Production changes
- **Absolute price change** → No change in incentives → Production unchanged

Relative vs Absolute Prices: An Example

A bicycle manufacturer faces a price increase:

Relative price change

Bicycle price: 500 € → 600 €
Other prices: unchanged

→ Profit per bicycle rises
→ Produce more bicycles!

This is the **micro** *story*

Absolute price change

Bicycle price: 500 € → 600 € (+20%)
Steel, wages, rent: all +20%

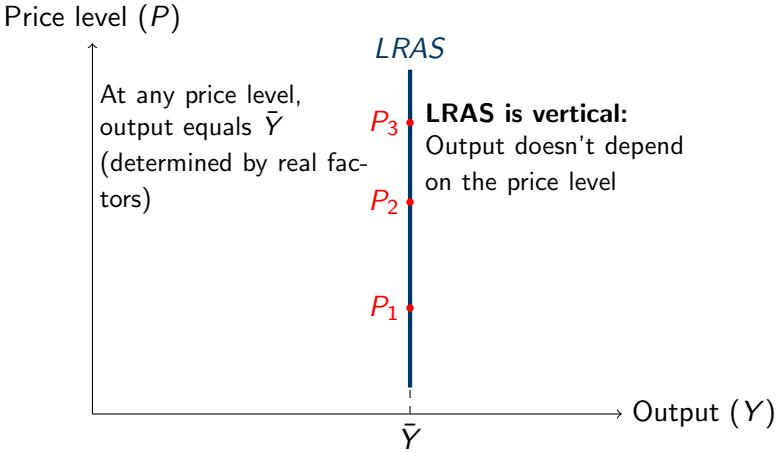
→ Profit per bicycle unchanged
→ No reason to change production

This is the **macro** *story (long run)*

In the long run: all prices adjust proportionally → **relative prices are constant**

If relative prices don't change, production doesn't change

Long-Run Aggregate Supply (LRAS)



$\bar{Y}$ is **potential GDP**: the output level when all resources are fully employed. Also called *natural rate of output*.

The Classical Dichotomy

Classical dichotomy: In the long run, real variables are determined independently of nominal variables.

Real variables
(determined by supply side)

- Output (Y)
- Employment (L)
- Real wages (W/P)
- Real interest rate (r)

Nominal variables
(determined by money supply)

- Price level (P)
- Nominal wages (W)
- Nominal interest rate (i)
- Inflation (π)

Aggregate demand is irrelevant for long-run production. Changes in AD only change P, not Y.

This changes in the short run, when prices are sticky (Topics 7-8).

The Production Function

What determines $\bar{Y}$?

We know the LRAS is vertical at $\bar{Y}$. But what makes $\bar{Y}$ large or small? The answer is the **aggregate production function**:

$$\bar{Y} = A \cdot F(L, K, H, N)$$

Inputs (factors of production):

- L Labor (number of workers, hours worked)
- K Physical capital (machinery, buildings, infrastructure)
- H Human capital (education, skills, training)
- N Natural resources (land, oil, minerals)
- A Technology (how efficiently we combine inputs)

This is the macro version of the production function from Intro I (ch. 13 of Mankiw & Taylor).

Key Properties of the Production Function

1. Constant returns to scale

If we double ALL inputs $\rightarrow$ output doubles

$$F(\lambda L, \lambda K, \lambda H, \lambda N) = \lambda F(L, K, H, N)$$

2. Diminishing marginal product

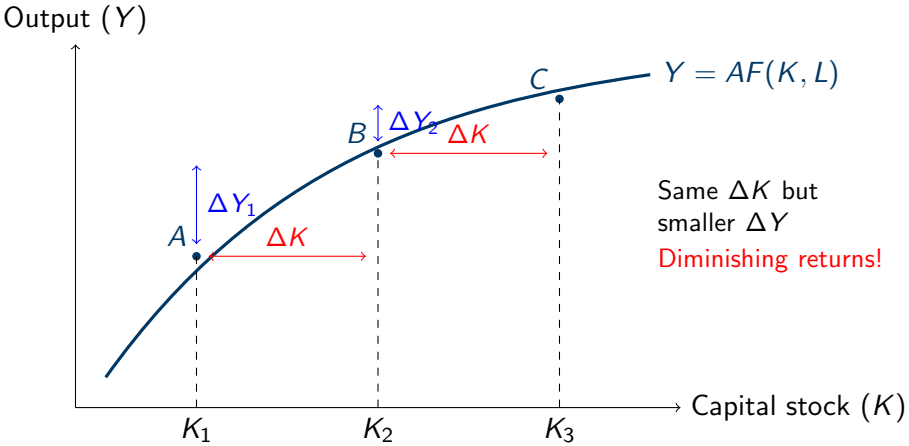
If we increase ONE input (holding others constant) $\rightarrow$ output increases, but at a decreasing rate

- **Marginal product of labor:** $\frac{\partial F}{\partial L} > 0$ but $\frac{\partial^2 F}{\partial L^2} < 0$
- **Marginal product of capital:** $\frac{\partial F}{\partial K} > 0$ but $\frac{\partial^2 F}{\partial K^2} < 0$

Why diminishing returns?

- Add workers to fixed capital $\rightarrow$ eventually they get in each other's way
- Add machines to fixed labor $\rightarrow$ some machines sit idle

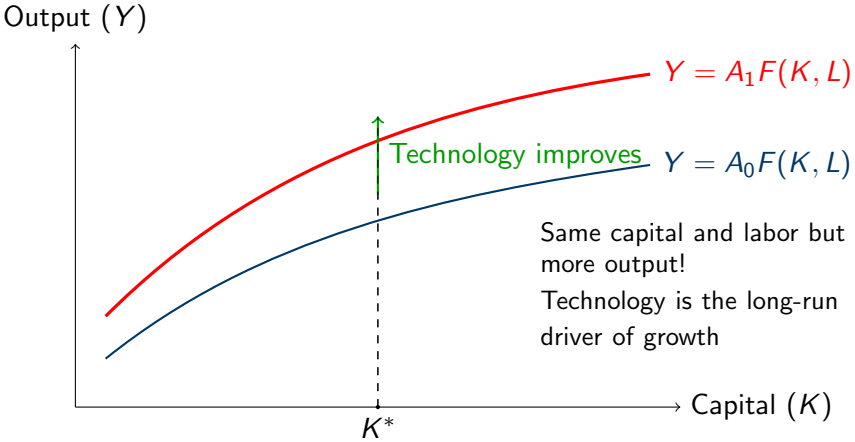
Visualizing the Production Function



Key insight: Each additional unit of capital adds less to output

Technology: The Game Changer

Technology (A) shifts the production function upward



What Determines Potential Output?

What determines $\bar{Y}$?

We know: $\bar{Y} = A \cdot F(K, L, H, N)$
Simplification: Focus on K and L: $\bar{Y} = A \cdot F(K, L)$

Key questions:

- What determines K and L?
- Do they depend on price level P?

If K and L do NOT depend on P $\rightarrow$ $\bar{Y}$ does NOT depend on P

This proves LRAS is vertical

Labor Demand: The Microeconomic Foundation

From Intro I: Firms hire by comparing benefit vs cost

Benefit: Value of Marginal Product

$$VMPL = P \times MPL$$

where MPL = extra output from one more worker

Cost: Nominal wage W

Optimal hiring rule:

$$\boxed{VMPL = W}$$

$$\boxed{P \times MPL = W}$$

Hire workers until the value they produce equals what they cost

From Nominal to Real Terms

Starting from: $P \times MPL = W$

Divide both sides by P:

$$\boxed{MPL = \frac{W}{P}}$$

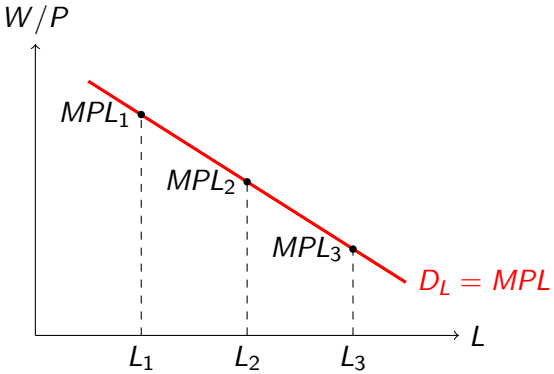
Interpretation:

- **Left side** (MPL): Physical productivity — technological fact
- **Right side** (W/P): **Real wage** — output cost of hiring

Firm's decision in real terms: Does this worker produce more output than the output I give up to pay them?

Since MPL is decreasing (diminishing returns) $\rightarrow$ labor demand slopes down

The Labor Demand Curve



Key property: Demand slopes down due to diminishing marginal product

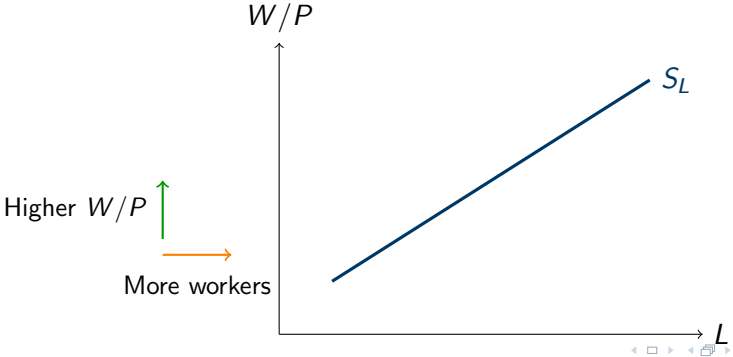
As we hire more workers, each additional worker produces less (holding K, A constant)

Labor Supply

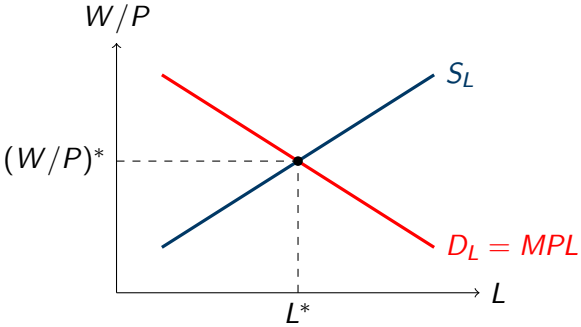
Workers face a trade-off: Work vs Leisure

Opportunity cost of leisure = Real wage W/P

- Higher $W/P \rightarrow$ More output sacrificed when not working
- $\rightarrow$ More incentive to work



Labor Market Equilibrium



Equilibrium where: $D_L = S_L$

This determines:

- Real wage: $(W/P)^*$
- Employment level: L^*

Key observation: Both axes measured in REAL terms

Why the Curves Don't Depend on P

Labor demand curve: $D_L = MPL$

- MPL = Physical productivity (technological)
- Depends on: K, L, A
- Does NOT depend on: P
- $\rightarrow$ Demand curve doesn't shift when P changes

Labor supply curve: Workers decide based on W/P

- Workers care about purchasing power (real wage)
- Supply function: $L^S = f(W/P)$
- Does NOT depend on P separately
- $\rightarrow$ Supply curve doesn't shift when P changes

Both curves stay put when price level changes!

What Happens When P Doubles?

Initial equilibrium: $P = 1$, $W = 10$, $W/P = 10$, $L^* = 100$

Price level doubles: P goes from 1 to 2

Step-by-step adjustment:

1. Initially: $W = 10$, $P = 2 \rightarrow W/P = 5$ (real wage fell!)
2. Firms want to hire more at lower real wage
3. Workers want to supply less at lower real wage
4. Excess demand for labor $\rightarrow$ W rises
5. Nominal wage adjusts to $W = 20$
6. New equilibrium: $W/P = 20/2 = 10$
7. Employment back to $L^* = 100$

Result: Nominal wage doubled, real wage unchanged, employment unchanged

Employment is independent of the price level!

Capital Demand: The Same Logic

Firms decide how much capital to rent:

Benefit: Value of Marginal Product of Capital

$$VMPK = P \times MPK$$

where MPK = extra output from one more unit of capital

Cost: Rental rate R

(The price of renting one unit of capital for one period)

Optimal rule:

$$\boxed{VMPK = R}$$

$$\boxed{P \times MPK = R}$$

Same logic as labor, but for machines and equipment

Capital in Real Terms

Starting from: $P \times MPK = R$

Divide both sides by P:

$$\boxed{MPK = \frac{R}{P}}$$

Interpretation:

- **Left side** (MPK): Physical productivity of capital
- **Right side** (R/P): **Real rental rate** — output cost of renting capital

Firm's decision: Does this machine produce more output than the output I give up to rent it?

Since MPK is decreasing (diminishing returns) $\rightarrow$ capital demand slopes down

The Capital Market: A Key Difference

Capital is a STOCK, not a flow:

Labor = Flow

- Workers choose hours each period
- Supply responds to W/P
- $\rightarrow$ Upward-sloping supply

Capital = Stock

- Machines already exist
- Can't "unbuild" factories
- All capital is used
- $\rightarrow$ **Vertical** supply

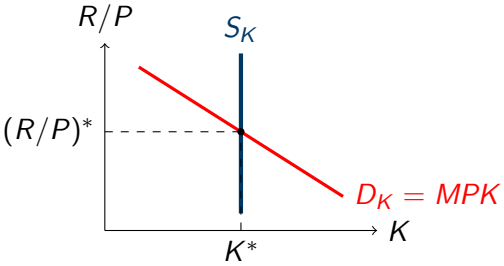
Capital stock evolves:

$$K_t = K_{t-1} + I_{t-1} - D_{t-1}$$

where I = gross investment, D = depreciation

At any moment in time, K is given (predetermined by past investment)

Capital Market Equilibrium



Equilibrium: All existing capital is used ($K^* = \bar{K}$)
Only the real rental rate $(R/P)^*$ adjusts to clear the market

When P doubles:

- Demand: $MPK = R/P$ (condition unchanged)
- Supply: Vertical at $\bar{K}$ (doesn't depend on anything)
- What adjusts? Nominal rental rate R doubles
- Result: R/P stays at $(R/P)^*$, and K^* unchanged

Why LRAS Is Vertical: The Complete Proof

Step 1: Labor market determines L^*

- Equilibrium: $MPL = W/P$ (both in real terms)
- When P changes: W adjusts proportionally, L^* unchanged

Step 2: Capital market determines K^*

- Equilibrium: $MPK = R/P$ (both in real terms)
- When P changes: R adjusts proportionally, K^* unchanged

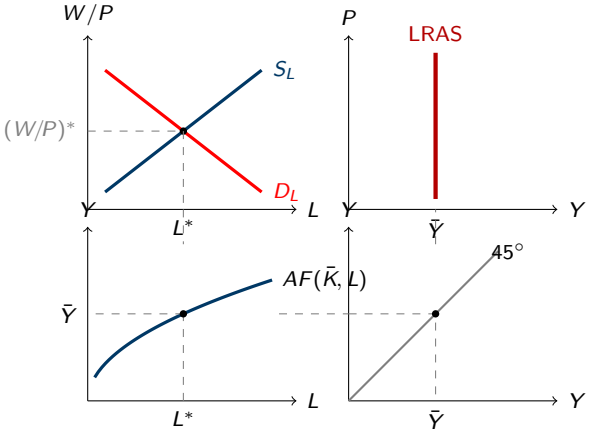
Step 3: Production function

- If K^* and L^* don't change, output doesn't change:

$$\bar{Y} = AF(K^*, L^*) \text{ is independent of } P$$

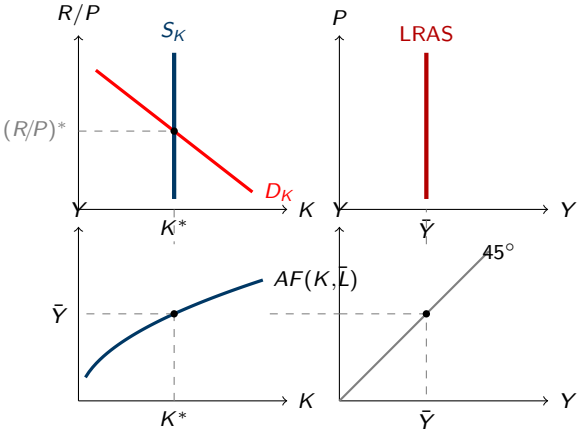
LRAS is vertical!

The Full Picture: How Labor Determines Output



L^* from the labor market $\downarrow$ production function $\rightarrow$ $\bar{Y}$ $\rightarrow$ 45° line $\uparrow$ LRAS vertical at $\bar{Y}$

The Full Picture: How Capital Determines Output



K^* from the capital market $\downarrow$ production function $\rightarrow$ $\bar{Y}$ $\rightarrow$ 45° line $\uparrow$ LRAS vertical at $\bar{Y}$

Can Factor Accumulation Sustain Growth?

We know how K^* and L^* are determined. But what happens when they grow?

Increasing capital or labor (K, L)**:**

- Raises the **level** of output (shifts LRAS right)
- But diminishing returns $\rightarrow$ each additional unit adds less
- The growth boost is **temporary**

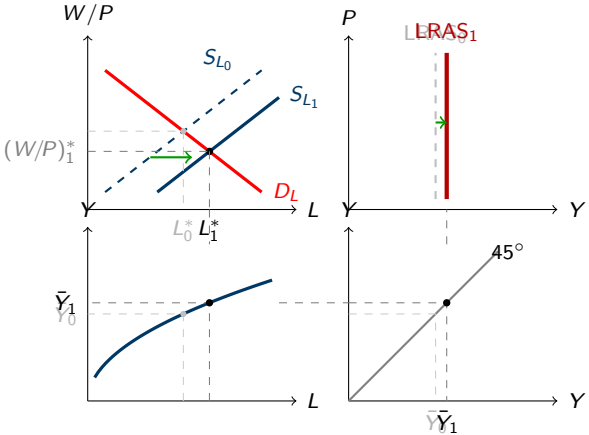
Improving technology (A)**:**

- Also shifts LRAS right, but is **not** subject to diminishing returns
- Can sustain **growth** indefinitely
- Key to long-run prosperity

Technology is the only source of sustained long-run growth

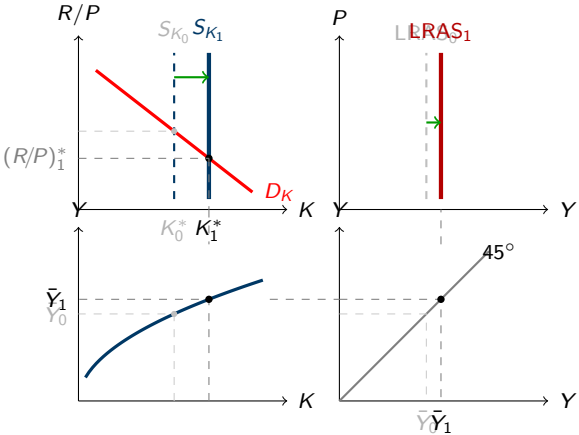
Let's see how each shock moves through the four-panel model...

Shock: Increase in Population ($\uparrow S_L$)



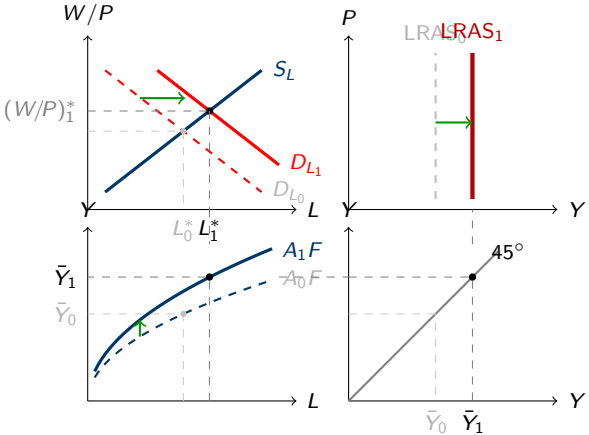
$\uparrow$ Population $\rightarrow$ S_L shifts right $\rightarrow \uparrow L^*$, $\downarrow (W/P)^* \rightarrow \uparrow \bar{Y} \rightarrow$ LRAS shifts right

Shock: Increase in Capital Stock ($\uparrow S_K$)



$\uparrow$ Investment $\rightarrow$ S_K shifts right $\rightarrow \uparrow K^*, \downarrow (R/P)^* \rightarrow \uparrow \bar{Y} \rightarrow$ LRAS shifts right

Shock: Technological Progress ($\uparrow A$) that affects MPL



$\uparrow A$: production fn shifts **up** + D_L shifts **right** $\rightarrow \uparrow L^*$, $\uparrow (W/P)^* \rightarrow$ **large** $\uparrow \bar{Y}$

The Rule of 70: Why Growth Rates Matter

We've seen that differences in technology, capital, or population growth shift the LRAS. But how much does a sustained difference matter over decades?

The Rule of 70: If a variable grows at $g\%$ per year, it doubles in approximately:

$$\text{Years to double} \approx \frac{70}{g}$$

Examples:

- At 1% growth: $70/1 = 70$ years to double
- At 2% growth: $70/2 = 35$ years to double
- At 3% growth: $70/3 \approx 23$ years to double
- At 7% growth: $70/7 = 10$ years to double

Small differences in sustained growth rates have enormous effects over time!

The Rule of 70: A Numerical Example

The Milton vs Janet Example:

Two economics graduates start at age 22, both earning 20,000 €/year:

- **Milton** lives in a slow-growth economy: 1% per year
- **Janet** lives in a faster-growth economy: 3% per year

After 40 years (at age 62):

$$\text{Milton's income} = 20{,}000 \times (1.01)^{40} = 29{,}777\,e$$
$$\text{Janet's income} = 20{,}000 \times (1.03)^{40} = 65{,}240\,e$$

A 2 pp difference in growth → Janet earns more than twice Milton!

Formula: $M = P(1+g)^n$ where P = initial value, g = rate, n = years

From Growth to Living Standards

What we've learned: Factor markets $\rightarrow K^*$, $L^* \rightarrow \bar{Y} = AF(K^*, L^*)$
Technology (A) is key for sustained growth.

But what determines LIVING STANDARDS?

The crucial insight:
Living standards depend on output **per person**, not total output.

- Country A: $\bar{Y} = \$1\text{T}$, 10M people $\rightarrow \$100,000$ per person
- Country B: $\bar{Y} = \$1\text{T}$, 100M people $\rightarrow \$10,000$ per person

We need to focus on Y/L (output per worker) GDP per capita

This is why PRODUCTIVITY matters

From Aggregate to Per-Worker Terms

Starting point: Aggregate production function

$$Y = A \cdot F(K, L, H, N)$$

Divide both sides by L:

$$\frac{Y}{L} = A \cdot \frac{F(K, L, H, N)}{L}$$

Using constant returns to scale:
Multiply all inputs by $1/L \rightarrow$ Output also multiplied by $1/L$

$$\frac{F(K, L, H, N)}{L} = F\left(\frac{K}{L}, 1, \frac{H}{L}, \frac{N}{L}\right)$$

The Per-Worker Production Function

Therefore:

$$\boxed{\frac{Y}{L} = A \cdot F\left(1, \frac{K}{L}, \frac{H}{L}, \frac{N}{L}\right)}$$

Output per worker depends on: capital per worker, human capital per worker, natural resources per worker, and technology.

This is PRODUCTIVITY:

$$\textbf{Productivity} = \frac{Y}{L}$$

Why it matters:

- Higher $Y/L \rightarrow$ Higher wages $\rightarrow$ Higher living standards
- Explains why some countries are rich and others poor

The Four Determinants of Productivity

$$\frac{Y}{L} = A \cdot F\left(1, \frac{K}{L}, \frac{H}{L}, \frac{N}{L}\right)$$

What makes workers more productive?

1. **Physical capital per worker** (K/L)
   Machines, equipment, infrastructure
2. **Human capital per worker** (H/L)
   Education, skills, training, health
3. **Natural resources per worker** (N/L)
   Land, minerals, energy
4. **Technological knowledge** (A)
   Better methods to produce output from inputs

Determinant 1: Physical Capital (K/L)

How to increase K/L:

Domestic

- Tax incentives for saving
- Public infrastructure
- Lower capital taxes

Foreign

- Foreign Direct Investment
- Open capital markets
- Stable institutions

Trade-off: Save more today $\rightarrow$ Consume less today, grow faster tomorrow

Example: Spain EU funds $\rightarrow$ Infrastructure $\rightarrow$ $K/L \uparrow$ $\rightarrow$ Productivity $\uparrow$

Determinant 2: Human Capital (H/L)

Why it matters:

- Educated workers understand complex tasks
- Trained workers make fewer mistakes
- Healthy workers are more productive

How to increase H/L:

Education

- Public schools
- Subsidized university
- Vocational training

Health

- Healthcare system
- Nutrition programs
- Disease prevention

Determinant 3: Natural Resources (N/L)

Impact on productivity:

- Resources **can help**: Norway, Saudi Arabia, Canada
- But **not necessary**: Japan, Singapore, Switzerland
- And **not sufficient**: Many poor resource-rich countries

Why less important than you'd think:

- Technology and human capital matter more
- Resources can be imported through trade
- "Resource curse": Can lead to poor institutions

Policy role: Limited — governments can't create resources

Natural resources help, but A and H/L are far more important

Determinant 4: Technology (A) - The Key!

Why technology is special:

- **NOT subject to diminishing returns**
- Can sustain growth indefinitely
- Public good $\rightarrow$ Benefits everyone

How to increase A:

Public R&D

- Universities, labs
- Basic research
- Defense, health, space

Private incentives

- R&D tax credits
- Patent protection
- Tech diffusion

Why government? Firms under-invest (can't capture all benefits)

Examples: Internet (DARPA), GPS (military), vaccines (NIH)

Technology = better use of inputs, whatever that means!

Can We Measure the Sources of Growth?

What we've established so far:

- Factor markets determine K^* and $L^* \rightarrow \bar{Y} = AF(K^*, L^*)$
- Factor accumulation (K, L) raises the level of output
- Technology (A) is the only source of sustained growth

But can we quantify how much of a country's growth comes from K, from L, and from A?

The problem:
With a generic $Y = AF(K, L)$, we can't decompose growth.
We need a **specific functional form** — one that lets us assign numbers to each source.

The Tool: The Cobb-Douglas Production Function

The most widely used production function in macroeconomics:

$$\boxed{Y = AK^{\alpha}L^{1-\alpha}}$$

Where:

- A = Total Factor Productivity (technology)
- K = Capital, L = Labor
- α = A parameter between 0 and 1

Good news: It satisfies the properties we already know:

- Constant returns to scale ✓
- Diminishing returns to each factor ✓

Named after mathematician Charles Cobb and economist Paul Douglas (1928), who found that this simple formula fits US production data remarkably well.

Why α and $1 - \alpha$?

You may wonder: Why not $Y = AK^{\alpha}L^{\beta}$ with two separate parameters?

Because we want constant returns to scale (CRS):
Double all inputs $\rightarrow$ Output doubles

$$A(2K)^{\alpha}(2L)^{\beta} = 2^{\alpha+\beta} \cdot AK^{\alpha}L^{\beta}$$

For this to equal $2Y$, we need $\alpha + \beta = 1$, i.e. $\beta = 1 - \alpha$.

So with CRS, a single parameter α governs everything:

- How much output responds to capital (α)
- How much output responds to labor ($1 - \alpha$)

This is what makes the model so parsimonious: one number captures the entire "balance" between capital and labor in production. But we still need to find its value...

The Calibration Problem

To use Cobb-Douglas for growth accounting, we need to know α.

$$Y = AK^{\alpha}L^{1-\alpha}$$

What is α?

- It controls how much output responds to capital vs labor
- It's the **elasticity of output with respect to capital**
- We need a number! Is it 0.2? 0.5? 0.8?

How can we pin down α empirically?

We could try to estimate α econometrically, but there's a much simpler path. It turns out α is directly observable in national accounts data. To see why, we need to derive the marginal products...

Marginal Products in Cobb-Douglas

We derive the marginal products to connect theory to factor markets — and from there, to observable data.

From $Y = AK^{\alpha}L^{1-\alpha}$ (full derivation in Appendix):

Marginal product of labor:

$$\boxed{MPL = \frac{\partial Y}{\partial L} = (1-\alpha)\frac{Y}{L}}$$

Marginal product of capital:

$$\boxed{MPK = \frac{\partial Y}{\partial K} = \alpha\frac{Y}{K}}$$

Read the MPL as: "the extra output from one more worker is $(1-\alpha)$ *times average output per worker." The exponent is the fraction — hence finding* α *is key.*

What Does This Mean for Factor Payments?

We connect Cobb-Douglas to what we learned earlier: firms pay each factor its marginal product.

Equilibrium conditions:

$$\frac{W}{P} = MPL = (1-\alpha)\frac{Y}{L} \qquad ; \qquad \frac{R}{P} = MPK = \alpha\frac{Y}{K}$$

Multiply each side by the quantity of the factor:

Total labor compensation: $\frac{W}{P} \cdot L = (1-\alpha)\frac{Y}{L} \cdot L = (1-\alpha) \cdot Y$

Total capital compensation: $\frac{R}{P} \cdot K = \alpha\frac{Y}{K} \cdot K = \alpha \cdot Y$

Workers collectively receive a fixed fraction $(1-\alpha)$ of total output. Capital owners get α. Always — regardless of how big the economy is.

From Total Payments to Factor Shares

From the previous slide:

$$(W/P) \cdot L = (1-\alpha) \cdot Y \qquad ; \qquad (R/P) \cdot K = \alpha \cdot Y$$

Why shares? We want what **proportion** of output goes to each factor — because that proportion is exactly α.

Divide both sides by Y:

$$\frac{(W/P) \cdot L}{Y} = 1-\alpha \qquad ; \qquad \frac{(R/P) \cdot K}{Y} = \alpha$$

In practice: National accounts (INE, Eurostat) report data in euros, not in real terms. Multiplying top and bottom by P gives us the same ratio in nominal terms:

$$\boxed{\frac{WL}{PY} = 1-\alpha} \qquad ; \qquad \boxed{\frac{RK}{PY} = \alpha}$$

WL = total wage bill (€), RK = capital income (€), PY = nominal GDP (€). These are all numbers you can look up directly.

The Elegant Result: α Is Directly Observable

$$\boxed{\text{Labor's share} = \frac{WL}{PY} = 1 - \alpha} \qquad \boxed{\text{Capital's share} = \frac{RK}{PY} = \alpha}$$

This is remarkable:

- α and $(1 - \alpha)$ are **constant** — independent of K, L, or Y
- They are **directly observable** in national accounts
- No econometric estimation needed!

How to read this:

- If labor's share of GDP $\approx 2/3 \rightarrow$ then $(1 - \alpha) \approx 2/3 \rightarrow \alpha \approx 1/3$
- That single number α is all we need for growth accounting

This is the payoff of choosing Cobb-Douglas: we don't need to estimate α — we just look up how GDP is split between wages and capital income.

Output Exhaustion: Why the Shares Add Up

Notice that:

$$\underbrace{(1-\alpha)\cdot Y}_{\text{Labor's total pay}} + \underbrace{\alpha\cdot Y}_{\text{Capital's total pay}} = Y$$

Paying each factor its marginal product **exactly exhausts** total output. Nothing left over, nothing missing.

This is Euler's Theorem:

$$\boxed{Y = MPL\cdot L + MPK\cdot K}$$

Why does this hold? Because of *constant returns to scale*. Under CRS, if you pay every input its marginal contribution, you distribute exactly 100% of output.

"Zero economic profits" doesn't mean firms earn nothing! It means all revenue goes to paying factors. No unexplained surplus — the shares cleanly decompose total income.

Empirical Evidence: The Shares Are Remarkably Stable

	Labor share $(1-\alpha)$	**Capital share** (α)
United States	≈ 0.64	≈ 0.36
Euro Area	≈ 0.62	≈ 0.38
Spain	≈ 0.60	≈ 0.40
OECD average	≈ 0.65	≈ 0.35

Key facts:

- Labor consistently gets about 60–65% of national income
- Shares are fairly stable over decades (though labor share has declined slightly since the 1980s — active research topic)
- **Standard calibration:** $\alpha \approx 1/3$

Sources: AMECO, Penn World Tables, OECD National Accounts

What Do These Shares Mean in Practice?

For Spain ($\alpha \approx 0.40$, labor share ≈ 0.60):

Of every 100 € of Spanish GDP:

$\approx$ 60 € goes to **workers** (wages, salaries, benefits)
$\approx$ 40 € goes to **capital owners** (profits, rents, interest)

This split is not a policy choice — it emerges from technology and competition. Kaldor (1961) identified stable factor shares as a "stylized fact" of growth. Cobb-Douglas predicts exactly this.

Now we have α. This means we can finally do what we set out to do: decompose economic growth into its sources.

The Growth Accounting Equation

This is the payoff of all our work. We have a function, we have α, we have data. Now we can decompose growth.

From Cobb-Douglas, taking logarithms and differentiating:

$$\boxed{\frac{\Delta Y}{Y} = \frac{\Delta A}{A} + \alpha \frac{\Delta K}{K} + (1-\alpha)\frac{\Delta L}{L}}$$

Interpretation:

$\frac{\Delta Y}{Y}$	= Output growth (observable)
$\alpha \frac{\Delta K}{K}$	= Capital's contribution (observable, weighted by α)
$(1-\alpha)\frac{\Delta L}{L}$	= Labor's contribution (observable, weighted by $1-\alpha$)
$\frac{\Delta A}{A}$	= TFP growth (the residual — not directly observable)

The weights are exactly the factor shares! Capital growth is weighted by $\alpha \approx 1/3$, labor by $2/3$. Not a coincidence — it's a direct consequence of Cobb-Douglas.

The Solow Residual

We observe $\Delta Y/Y$, $\Delta K/K$, $\Delta L/L$, and we know α.
Technology is what's left over:

$$\boxed{\frac{\Delta A}{A} = \frac{\Delta Y}{Y} - \alpha\frac{\Delta K}{K} - (1-\alpha)\frac{\Delta L}{L}}$$

This is the **Solow Residual** (Robert Solow, 1957).
What does TFP capture?

- Technological innovation
- Better management practices
- Institutional improvements
- More efficient resource allocation

Often called "a measure of our ignorance" — everything that drives growth beyond adding more workers and machines. Yet this "ignorance" is the most important driver of prosperity.

Numerical Example: Spain 2010–2020

Observed data:

Output growth:	$\Delta Y/Y = 2.0\%$
Capital growth:	$\Delta K/K = 2.5\%$
Labor growth:	$\Delta L/L = 0.5\%$
Capital share:	$\alpha = 1/3$

Step 1: Capital's contribution

$$\alpha \frac{\Delta K}{K} = \frac{1}{3} \times 2.5\% = 0.83 \text{ pp}$$

Step 2: Labor's contribution

$$(1-\alpha)\frac{\Delta L}{L} = \frac{2}{3} \times 0.5\% = 0.33 \text{ pp}$$

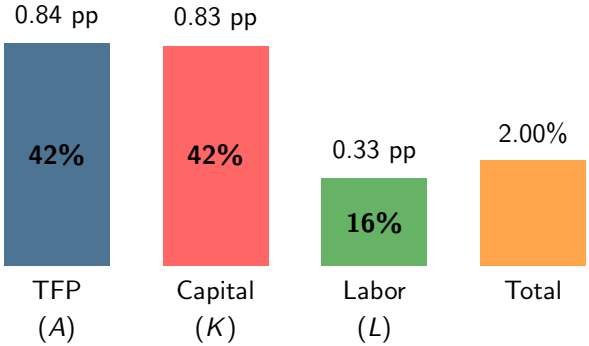
Step 3: TFP (residual)

$$\frac{\Delta A}{A} = 2.0 - 0.83 - 0.33 = \boxed{0.84 \text{ pp}}$$

All we needed: the formula, α from national accounts, and three observable growth rates.

Interpreting the Decomposition

Spain's 2% growth came from:



Key insight: Technology (TFP) explains as much as capital accumulation. And TFP is the only source of *sustained* growth (no diminishing returns).

What Growth Accounting Reveals

Growth accounting is a powerful diagnostic tool:

High TFP growth

- Innovation-driven economy
- Good institutions
- Efficient resource allocation
- Example: US, Nordic countries

Low TFP growth

- Growth only from "more inputs"
- Unsustainable (diminishing returns)
- May signal institutional problems
- Example: USSR in the 1960s–80s

Policy implication:
A country that grows only by accumulating K and L will eventually stagnate. Sustained prosperity requires investing in A: R&D, education, institutions, openness.

This is why Mankiw's Principle 8 matters: a country's standard of living depends on its ability to produce — and that ability depends, above all, on technology.

Back to the Fundamentals: Mankiw's Principle 8

"A country's standard of living depends on its ability to produce goods and services"

What we've learned in this topic:

1. **Production capacity determines living standards**
   - $\bar{Y} = A \cdot F(K, L, H, N)$ determines what we CAN produce
   - Higher Y/L (productivity) $\rightarrow$ Higher real wages $\rightarrow$ Better lives
2. **Long-run aggregate supply is independent of prices**
   - LRAS is vertical: determined by real factors (A, K, L), not by P
   - Aggregate demand doesn't affect $\bar{Y}$ in the long run

The Long Run: Two Interpretations

The concept of "long run" has two related meanings:

1. **Notional (equilibrium concept):**
   - All prices and wages are fully flexible
   - All markets clear (supply = demand)
   - Economy operates at full capacity ($Y = \bar{Y}$)
   - *This is the state we analyze, regardless of time*

2. **Temporal (time horizon):**
   - Enough time for capital stock to grow (investment $\rightarrow$ K $\uparrow$)
   - Enough time for population to change (L $\uparrow$)
   - Enough time for technology to improve (A $\uparrow$)
   - *Typically: decades, not months or years*

Both meanings coincide: In the long run, the economy reaches full employment equilibrium, and that's when factors can grow.

Summary: The Determinants of Prosperity

Economic growth depends on:

1. **Factor accumulation:**
   - More capital per worker ($K/L \uparrow$)
   - More human capital per worker ($H/L \uparrow$)
   - But: diminishing returns $\rightarrow$ Cannot sustain growth alone
2. **Technological progress (A):**
   - NOT subject to diminishing returns
   - The ONLY source of sustained long-run growth
   - Shifts production function: more output from same inputs

Income distribution:

- Factors are paid their marginal product: $W/P = MPL$, $R/P = MPK$
- With constant returns to scale: factor payments exhaust output
- $Y = MPL \cdot L + MPK \cdot K$ (Euler's Theorem)

Appendix: Derivation of Marginal Products

Starting point: Cobb-Douglas production function

$$Y = AK^{\alpha}L^{1-\alpha}$$

We will derive:

1. Marginal product of labor (MPL)
2. Marginal product of capital (MPK)

Tools we'll use:

- Power rule: $\frac{d}{dx}x^n = nx^{n-1}$
- Partial derivative: treat other variables as constants
- Properties of exponents: $x^a \cdot x^b = x^{a+b}$, $\frac{x^a}{x^b} = x^{a-b}$

Appendix: Deriving MPL (Steps 1-2)

Marginal Product of Labor: $MPL = \frac{\partial Y}{\partial L}$

$$\frac{\partial Y}{\partial L} = \frac{\partial}{\partial L}\left[AK^{\alpha}L^{1-\alpha}\right]$$

Step 1: Take out constants (A and K^{α} don't depend on L)

$$= AK^{\alpha} \cdot \frac{\partial}{\partial L}\left[L^{1-\alpha}\right]$$

Step 2: Apply power rule: $\frac{d}{dx}x^n = nx^{n-1}$

$$= AK^{\alpha} \cdot (1-\alpha) \cdot L^{(1-\alpha)-1}$$

Appendix: Deriving MPL (Step 3)

Continuing from: $AK^{\alpha} \cdot (1-\alpha) \cdot L^{(1-\alpha)-1}$

Step 3: Simplify exponent: $(1-\alpha)-1 = 1-\alpha-1 = -\alpha$

$$= AK^{\alpha} \cdot (1-\alpha) \cdot L^{-\alpha}$$

Reordering:

$$= (1-\alpha) \cdot AK^{\alpha}L^{-\alpha}$$

Appendix: Deriving MPL (Step 4)

Continuing from: $(1-\alpha) \cdot AK^{\alpha}L^{-\alpha}$

Step 4: Rewrite negative exponent using: $L^{-\alpha} = \frac{1}{L^{\alpha}}$

$$= (1-\alpha) \cdot AK^{\alpha} \cdot \frac{1}{L^{\alpha}}$$

$$= (1-\alpha) \cdot \frac{AK^{\alpha}}{L^{\alpha}}$$

Appendix: Deriving MPL (Step 5)

Continuing from: $(1-\alpha) \cdot \frac{AK^{\alpha}}{L^{\alpha}}$

Step 5: Multiply numerator and denominator by $L^{1-\alpha}$

$$= (1-\alpha) \cdot \frac{AK^{\alpha} \cdot L^{1-\alpha}}{L^{\alpha} \cdot L^{1-\alpha}}$$

Why? To get Y in the numerator and simplify the denominator.

Appendix: Deriving MPL (Step 6)

Continuing from: $(1-\alpha) \cdot \frac{AK^{\alpha}L^{1-\alpha}}{L^{\alpha} \cdot L^{1-\alpha}}$

Step 6: Simplify denominator using: $L^a \cdot L^b = L^{a+b}$

$$L^{\alpha} \cdot L^{1-\alpha} = L^{\alpha+(1-\alpha)} = L^1 = L$$

Therefore:

$$= (1-\alpha) \cdot \frac{AK^{\alpha}L^{1-\alpha}}{L}$$

Appendix: Deriving MPL (Final Step)

Continuing from: $(1-\alpha) \cdot \frac{AK^{\alpha}L^{1-\alpha}}{L}$

Step 7: Recognize that $AK^{\alpha}L^{1-\alpha} = Y$ (our original production function!)

$$= (1-\alpha) \cdot \frac{Y}{L}$$

$$\boxed{MPL = (1-\alpha)\frac{Y}{L}}$$

Interpretation:

- MPL is proportional to average product of labor (Y/L)
- The proportion is labor's share in production $(1-\alpha)$
- Typical value: $(1-\alpha) \approx 0.7$ (labor gets 70% of output)

Appendix: Deriving MPK (Steps 1-2)

Marginal Product of Capital: $MPK = \frac{\partial Y}{\partial K}$

$$\frac{\partial Y}{\partial K} = \frac{\partial}{\partial K}\left[AK^{\alpha}L^{1-\alpha}\right]$$

Step 1: Take out constants (A and $L^{1-\alpha}$ don't depend on K)

$$= AL^{1-\alpha} \cdot \frac{\partial}{\partial K}\left[K^{\alpha}\right]$$

Step 2: Apply power rule: $\frac{d}{dx}x^n = nx^{n-1}$

$$= AL^{1-\alpha} \cdot \alpha \cdot K^{\alpha-1}$$

Appendix: Deriving MPK (Step 3)

Continuing from: $AL^{1-\alpha} \cdot \alpha \cdot K^{\alpha-1}$

Step 3: Reorder

$$= \alpha \cdot AL^{1-\alpha}K^{\alpha-1}$$

$$= \alpha \cdot AK^{\alpha-1}L^{1-\alpha}$$

Appendix: Deriving MPK (Step 4)

Continuing from: $\alpha \cdot AK^{\alpha-1}L^{1-\alpha}$

Step 4: Rewrite exponent using: $K^{\alpha-1} = K^{\alpha} \cdot K^{-1} = \frac{K^{\alpha}}{K}$

$$= \alpha \cdot A \cdot \frac{K^{\alpha}}{K} \cdot L^{1-\alpha}$$

$$= \alpha \cdot \frac{AK^{\alpha}L^{1-\alpha}}{K}$$

Appendix: Deriving MPK (Final Step)

Continuing from: $\alpha \cdot \frac{AK^{\alpha}L^{1-\alpha}}{K}$

Step 5: Recognize that $AK^{\alpha}L^{1-\alpha} = Y$

$$= \alpha \cdot \frac{Y}{K}$$

$$\boxed{MPK = \alpha \frac{Y}{K}}$$

Interpretation:

- MPK is proportional to average product of capital (Y/K)
- The proportion is capital's share in production α
- Typical value: $\alpha \approx 0.3$ (capital gets 30% of output)

Appendix: Summary of Marginal Products

Final results for Cobb-Douglas $Y = AK^{\alpha}L^{1-\alpha}$:

Marginal Product of Labor	Marginal Product of Capital
$MPL = \frac{\partial Y}{\partial L}$	$MPK = \frac{\partial Y}{\partial K}$
$= (1-\alpha)\frac{Y}{L}$	$= \alpha\frac{Y}{K}$

Key properties:

1. **Diminishing returns:** Both MPL and MPK decrease as L or K increase (holding the other constant)
2. **Constant shares:** The exponents α and $(1-\alpha)$ are also the factor shares in income
3. **Easy to compute:** Just need to know Y, K, L, and α — no need to differentiate each time!

Appendix: Diminishing Returns to Labor

Goal: Show that $\frac{\partial^2 Y}{\partial L^2} < 0$ (MPL decreases as L increases)

Starting from: $MPL = (1-\alpha) \cdot AK^{\alpha} L^{-\alpha}$

Take second derivative:

$$\frac{\partial^2 Y}{\partial L^2} = \frac{\partial MPL}{\partial L}$$

$$= \frac{\partial}{\partial L}\left[(1-\alpha) \cdot AK^{\alpha} L^{-\alpha}\right]$$

Step 1: Constants $(1-\alpha)$, A, K^{α} don't depend on L:

$$= (1-\alpha) \cdot AK^{\alpha} \cdot \frac{\partial}{\partial L}\left[L^{-\alpha}\right]$$

Appendix: Diminishing Returns to Labor (cont.)

Continuing: $(1-\alpha) \cdot AK^{\alpha} \cdot \frac{\partial}{\partial L}\left[L^{-\alpha}\right]$

Step 2: Apply power rule: $\frac{d}{dx}x^n = nx^{n-1}$

$$= (1-\alpha) \cdot AK^{\alpha} \cdot (-\alpha) \cdot L^{-\alpha-1}$$

$$= -(1-\alpha)\alpha \cdot AK^{\alpha}L^{-\alpha-1}$$

Sign analysis:

- $(1-\alpha) > 0$, $\alpha > 0$, $A > 0$, $K^{\alpha} > 0$, $L^{-\alpha-1} > 0$
- Negative sign in front $\rightarrow$ Whole expression is negative

$$\boxed{\frac{\partial^2 Y}{\partial L^2} < 0} \quad \checkmark$$

MPL decreases as L increases $\rightarrow$ Diminishing returns to labor!

Appendix: Diminishing Returns to Capital

Goal: Show that $\frac{\partial^2 Y}{\partial K^2} < 0$ (MPK decreases as K increases)

Starting from: $MPK = \alpha \cdot AL^{1-\alpha}K^{\alpha-1}$

Take second derivative:

$$\frac{\partial^2 Y}{\partial K^2} = \frac{\partial MPK}{\partial K}$$

$$= \frac{\partial}{\partial K}\left[\alpha \cdot AL^{1-\alpha}K^{\alpha-1}\right]$$

Step 1: Constants α, A, $L^{1-\alpha}$ don't depend on K:

$$= \alpha \cdot AL^{1-\alpha} \cdot \frac{\partial}{\partial K}\left[K^{\alpha-1}\right]$$

Appendix: Diminishing Returns to Capital (cont.)

Continuing: $\alpha \cdot AL^{1-\alpha} \cdot \frac{\partial}{\partial K}\left[K^{\alpha-1}\right]$

Step 2: Apply power rule:

$$= \alpha \cdot AL^{1-\alpha} \cdot (\alpha - 1) \cdot K^{\alpha-2}$$

$$= \alpha(\alpha - 1) \cdot AL^{1-\alpha}K^{\alpha-2}$$

Sign analysis:

- $\alpha > 0$, $(\alpha - 1) < 0$ (since $0 < \alpha < 1$), $A > 0$, $L^{1-\alpha} > 0$, $K^{\alpha-2} > 0$
- Product: $(+) \times (-) \times (+) \times (+) \times (+) = (-)$

$$\boxed{\frac{\partial^2 Y}{\partial K^2} < 0} \quad \checkmark$$

MPK decreases as K increases → Diminishing returns to capital!